

“PGM — Probabilistic Graphical Models” Exam

(29/01/2025) 2h00. R. EMONET

Student name:

(write in the most readable manner, give concise and clear explanations, do not copy the question itself)

..... (horizontal lines like this one separate mostly-unrelated exercises)

Exercise 1

Given the following generative story:



We observe 100 integer values (x), each being supposed to be obtained as follows:

- we first flip a coin (always the same) to get a value c (we consider that “Tails” is 1 and “Heads” is 2).

- if c is 1, we then throw a 6-sided dice to get the integer x (from 1 to 6)

- if c is 2, we then throw a 10-sided dice to get the integer x (from 1 to 10)

NB: there are two dices and one coin, and they are ****not necessarily fair**** (their outcome are not equiprobable).

Using the plate notation, what is the corresponding graphical representation?

Exercise 2...

What are the random variables in this problem? how many are there? are they observed, hidden, other?

Exercise 3...

How many parameters are required to specify this model?

Exercise 4...

Suppose a trustworthy expert gave you the parameters by telling you “the probability of each outcome is proportional to the outcome, i.e., for the 10-sided dice, throwing 10 is $\frac{10}{7}$ more probable than throwing 7, and remember that Tails is 1 and Heads is 2, so for we have $\frac{1}{3}$ chance of getting Tails and $\frac{2}{3}$ chance of getting Heads.”

How do you write and compute (if there is sufficient information) the probability that the last integer x_{100} is equal to 5, given that we flipped a Tails for this integer?

Exercise 5...

Similarly, how do you write and compute (if there is sufficient information) the probability that the last integer x_{100} is equal to 5?

Exercise 6...

Similarly, how do you write and compute (if there is sufficient information) the probability that we flipped “Tails” (for this integer) given that the last integer x_{100} is equal to 5?

Exercise 7



A study has been carried out on a thousand runners. For simplicity, only binarized information have been reported. For each runner, we collected: (F) whether he/she finished the goal race (0=no, 1=yes), (V) whether he/she had a low or high weekly training volume (0=less than 50km, 1=more than 50km), and (T) whether the race was on road or on trails (0=road, 1=trail). The results of the survey have been summarized in the provided table.

Finish?	Volume	Type	count
no	low	road	100
no	low	trail	40
no	high	road	10
no	high	trail	200
yes	low	road	300
yes	low	trail	10
yes	high	road	90
yes	high	trail	250

People do wonder if there is a gain to be expected by training with high volume. To measure this **gain** we use the ratio of two probabilities: the probability of finishing given a high training load, vs the probability of finishing given a low training load. If this ratio is greater than 1, a higher training volume improves chances of finishing. This ratio goes from 0 to infinity (side note: we sometimes take the log of such ratio, to have a “symmetric” range, ranging from minus infinity and to positive infinity).

Professor L. Azy got access to this data and analyzed it. You need to reproduce his analysis.

...

(compute) By using the provided counts, what is the formula and an estimate/approximation of the probabilities of interest and the gain (their ratio)?

Exercise 8...

(interpret) What conclusion L. Azy could draw about the effect of training with high volume?

Exercise 9...

Now K. Ipchoge, a road runner, obtains the data and wants to know precisely if he should train with high volume.

(compute) What is the gain (ratio) that you obtain by repeating his analysis?

Exercise 10...

(interpret) What would K. Ipchoge conclude about whether he should train with high volume?

Exercise 11...

Similarly F. Orsberg, a trail runner, obtains the data and wants to know precisely if she should train with high volume.

(compute) What is the gain (ratio) that you obtain by repeating her analysis?

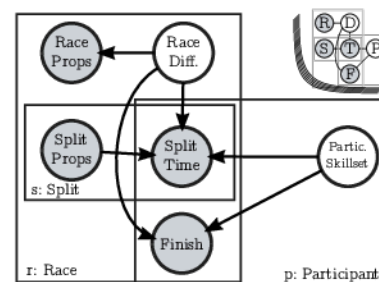
Exercise 12...

(interpret) What would F. Orsberg conclude about whether he should train with high volume?

Exercise 13...

There might seem like there is an incoherence between the 3 computed gains. According to you, what is the right decision/interpretation?

We consider a Bayesian Network for the problem of modeling race difficulties and participant skills. The top-right corner is a proposal of single-letter variable names. F is typically a boolean, R a set of features (total distance, total dpos, etc.), S a set of split features (distance, pos, alt, etc.).



(many of the questions can be answered even if you skip some of the previous ones)

Exercise 14

In this Bayesian network can we say that the "D" and the "P" are independent, given the "T"? Then give a proof.

Exercise 15

Write a generative story/process corresponding to this Bayesian Network. Where needed, specify/suggest distributions that you think make sense. Provide any details and modifications (if any) you think might be necessary.

Exercise 16

Explain the principle of a Gibbs sampler in this context, what it requires you to derive (do not provide the full actual derivations) and how it is used in the algorithm.

Exercise 17

Explain what would a variational inference approach mean in this context, what it requires you to derive (do not provide the full actual derivations) and how it is used in the algorithm.

Exercise 18

Explain how a Variational Auto-Encoder approach could be designed in this context, and if meaningful, how it relates to Gibbs sampling and/or variational inference.

Exercise 19

Explain what is doubly-stochastic variational inference in this context.

Exercise 20

Explain what are amortization and the reparametrization trick in this context?

Exercise 21

If you have acquired an important knowledge/skill/insight in this course+project that you cannot exploit/illustrate in the above exercises, please explain it.